

Assignment # 02: Practicing Control-Flow (Decision-Making Structures and Loops)

Objective:

To help you understand and apply control-flow in Python Programming.

- Use **if**, **if-else** and **if-elif-else** statements to make decisions in program.
- Practice using **for** and **while** loops to repeat actions based on conditions.
- Learn to combine conditional statements and loop to solve real-world problems.
- Develop logical thinking by implementing different control-flow structures.

Part 1: Decision-Making Structures

1. Write a program that checks if a given number is positive, negative or zero.
2. Write a program to find the larger of the two numbers.
3. Write a program to find the largest of the three numbers.
4. Write a program to check if the given string "Mass" is present in the string "Saylani Mass IT". If the string is present then display the message "string found".
5. Write a program to take user's age as input and display whether he is minor, adult or senior citizen:
 - a. Minor age is less than 18.
 - b. Adult age is greater than 18 and less than 60
 - c. Senior citizen age is greater than 60
6. Write a program to take integer as input and check if it is even or odd.
7. Write a program to take two numbers and an operator (/,x,+, -) as input and perform the corresponding operation.
8. Write a program to take number as input and check if it lies in the range i.e. 20(inclusive) – 40 (exclusive)
9. Write a program that asks for an integer number and checks if it is divisible by 2, 3, or both.
10. Write a program to take user's score as input and determine if they pass or fail (pass if score is above 60, otherwise fails.)
11. Write a program that evaluates if an input number is prime.
12. Write a program that takes a temperature in Celsius and checks if it is freezing, moderate or hot.
 - a. Freezing temperature is below 0.
 - b. Moderate temperature is greater than 0 and less than 26.
 - c. Hot temperature is above 26.

Part-2: Loops

1. Write a program to print numbers from 1 to 30 using for loop.
2. Write a program to print even numbers from 1 to 50 using while loop.
3. Write a program to print multiplication table of a given number.
4. Write a program to print odd numbers from 1 to 50 using while loop.
5. Write a program to calculate the sum of all numbers from 1 to 50 using for loop.
6. Write a program to print each character of a string.
7. Write a program to compute the factorial of a number using while loop.
8. Write a program to print numbers from 10 down to 1.
[Hint: first check the sequence generated by range function on IDLE using this command i.e. **list(range(10, 0, -1))**]
9. Write a program that continues to ask user to input password until the correct password is entered.
10. Write a program to print the square of each number from 1 to 10 using for loop.
11. Write a program to calculate the sum of even and odd numbers and print them. (numbers from 1 to 50)

Bonus Challenge:

Write a program to print the first 10 Fibonacci numbers using for loop.

Hint: First 10 Fibonacci numbers are → 0,1,1,2,3,5,8,13,21,34